

COMICCRAFT

DOCUMENT 20: README (Project Summary)

■ ComicCraft - AI Comic Story Creator using Gemini Models

Transform your ideas into stunning comics with the power of AI.

ComicCraft is a Generative AI-powered web application that lets anyone create professional 5-panel comic stories from a simple text prompt. No drawing skills needed. No expensive software required. Just your imagination.

■ Features

- Smart Input Form - Enter your story prompt, character name, setting, tone, and art style
- Gemini AI Story Engine - Dual-model architecture: Flash for outlines, Pro for rich narration
- AI Art Generation - Stable Diffusion / FLUX creates beautiful panel illustrations
- Interactive Preview - Browse your comic panel-by-panel with narration and dialogue
- PDF Export - Download professional multi-page PDF comics with Unicode support
- Mock Fallback - Works without API keys for testing and development

■■ Tech Stack

COMPONENT	TECHNOLOGY
Frontend	HTML5, CSS3, JavaScript, Jinja2
Backend	Python FastAPI + Uvicorn
AI Models	Google Gemini 1.5 Flash & 1.5 Pro
Image Gen	Stable Diffusion / FLUX.1-schnell
PDF Export	FPDF2 (fpdf2)
Storage	File system (static/)

■ Quick Start

```
git clone https://github.com/roherror404/ComicCraft.git
cd ComicCraft
python -m venv venv
venv\Scripts\activate # Windows
pip install -r requirements.txt
cp .env.example .env # Add your GEMINI_API_KEY
uvicorn app.main:app --reload
Open http://127.0.0.1:8000 in your browser.
```

■ Project Structure

```
ComicCraft/
  ■■■ app/                # Main application code
  ■   ■■■ main.py         # FastAPI entry point
  ■   ■■■ routes.py       # HTTP route handlers
  ■   ■■■ gemini_flash.py  # Outline generation
  ■   ■■■ gemini_pro.py    # Story generation
```

```
■ ■■■ image_generator.py # AI image creation
■ ■■■ layout_builder.py # Comic layout assembly
■ ■■■ exporters.py      # PDF export
■ ■■■ pipeline.py       # Generation pipeline
■■■ templates/          # Jinja2 HTML templates
■■■ static/             # CSS, JS, generated files
■■■ tests/              # Unit tests
■■■ .env.example        # Environment template
■■■ requirements.txt    # Python dependencies
```

■ How It Works

You input → Story prompt, character name, setting, tone, art style

Gemini Flash → Generates a structured 5-panel comic outline

Gemini Pro → Expands with rich narration and dialogue

Stable Diffusion → Creates panel illustrations

Layout Builder → Assembles everything into comic format

FPDF2 → Exports as a professional multi-page PDF

■ Performance

Text generation: ~1.2 seconds avg

Full comic (with images): ~15-25 seconds (GPU dependent)

PDF export: ~1-2 seconds

Mock fallback: ~1-2 seconds

■ Future Roadmap

User accounts & comic history

Multi-language support

Image-to-comic conversion

Social sharing & community gallery

More art styles & customization